

Q.1a) How we can distinguish between the message to be recorded in the snapshot from those not to be recorded?

b) Describe the snapshot algorithm and explain how it records consistent cuts.

3+7=10 [CO2]

Q.2a) Explain with an example how the presence of a cycle in a resource allocation graph is a necessary but not a sufficient condition for a deadlock to exist.

b) Why is the deadlock avoidance strategy never used in distributed systems for handling deadlocks?

c) Show that, in the AND model, false deadlocks can occur due to deadlock resolution in distributed systems. Can something be done about it or they are bound to happen?

d) What are the main advantages of a Probe based algorithm over a WFG based distributed deadlock detection algorithm.

3+2+3+2 = 10 [CO4]

Q.3 Consider a distributed system with three processes P1, P2 and P3 and three resources R1, R2 and R3. The system uses C-M-H algorithm for detecting deadlocks. The processes and resources are connected as follows: P1 holds R1 and requests R2, P2 holds R2 and requests R3, P3 holds R3 and requests R1.

(a) Explain how probes are propagated in the system, starting from P1

(b) Show the sequence of probe messages exchanged during the detection of a deadlock ?

(c) Does the system have a deadlock? Justify your answer based on the probe messages.

(d) What happens if P3 releases R3 before P1 sends its probe?

(e) How does C-M-H algorithm ensure correctness and avoid false positives in such scenarios?

5X2=10 [CO4]